

DELTA COUNTY AP, MI, USA

WMO: 726480

Lat: 45.733N

Lon: 87.083W

Elev: 181

StdP: 99.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-19.8	-28.0	0.3	-20.2	-25.7	0.4	-19.1	10.6	-7.5	9.2	-7.7	2.8	270	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	27.9	20.8	27.0	20.4	25.1	19.5	22.8	26.4	21.7	25.1	20.6	23.9	5.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.7	25.4	20.8	15.8	24.4	19.1	14.2	22.8	68.2	26.8	64.0	25.3	60.2	24.3	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	-26.2	31.4	3.3	1.8	-28.6	32.7	-30.5	33.8	-32.4	34.8	-34.8	36.1		
(5)			-26.9	24.4	3.3	1.3	-29.2	25.3	-31.1	26.1	-33.0	26.8	-35.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-8.1	-7.4	-2.4	3.8	10.4	16.0	18.9	18.2	14.4	7.9	1.5	-4.4		
	DBStd	10.57	6.00	6.14	5.57	3.99	4.00	3.72	3.30	3.07	4.00	4.25	4.73	5.38		
	HDD10.0	2469	561	488	384	188	43	2	0	0	8	94	256	445		
	HDD18.3	4687	819	721	642	435	246	88	34	40	129	324	504	703		
	CDD10.0	942	0	0	1	2	57	183	274	254	139	29	2	0		
	CDD18.3	118	0	0	0	0	2	19	51	36	10	1	0	0		
	CDH23.3	785	0	0	0	1	18	133	377	214	41	1	0	0		
	CDH26.7	128	0	0	0	0	3	24	74	23	3	0	0	0		
(14) Wind	WSAvg	3.5	3.7	3.6	3.6	3.8	3.5	3.2	3.1	3.1	3.3	3.7	3.8	3.7		
(15) Precipitation	PrecAvg	767	36	26	47	64	83	85	85	79	90	77	56	39		
	PrecMax	958	96	49	107	137	157	182	205	114	203	140	116	73		
	PrecMin	619	10	1	11	19	34	36	20	25	34	23	20	4		
	PrecStd	94	21	13	28	28	35	34	41	23	38	37	26	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.7	13.7	18.0	26.1	29.1	31.2	28.9	27.0	21.3	14.9	8.7		
		MCWB	3.0	3.8	8.6	10.3	16.6	19.9	21.8	21.8	20.3	17.2	10.8	7.6		
	2%	DB	2.9	3.9	9.1	14.2	22.3	27.0	28.6	27.3	24.1	18.7	12.2	6.2		
		MCWB	1.4	1.9	5.5	8.1	15.1	19.5	21.6	20.9	19.0	15.7	9.6	4.4		
	5%	DB	2.0	2.4	7.3	12.6	19.7	24.6	27.3	26.0	22.8	17.2	10.4	4.0		
		MCWB	1.0	0.8	4.0	7.2	13.8	18.7	20.9	20.1	18.7	13.9	8.2	2.4		
	10%	DB	0.8	1.2	5.0	11.1	17.6	22.7	25.8	24.1	21.7	15.3	8.7	2.7		
		MCWB	-0.1	-0.2	2.2	6.2	12.5	17.2	20.0	19.2	18.0	12.6	6.9	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	4.6	9.8	12.3	18.8	22.8	24.5	23.5	22.2	18.6	11.9	7.8		
		MCDB	4.9	6.1	13.0	16.1	22.7	27.6	28.4	27.0	25.3	19.8	14.3	8.2		
	2%	WB	1.9	2.4	6.4	9.8	16.5	20.7	23.2	22.4	20.7	16.4	10.1	4.8		
		MCDB	2.8	3.8	8.5	13.4	20.4	25.1	26.8	25.8	22.9	17.9	11.7	6.2		
	5%	WB	0.7	0.9	4.2	7.9	14.8	19.4	22.0	21.3	19.6	14.4	8.3	3.1		
		MCDB	1.6	2.2	6.9	11.5	18.8	23.3	25.5	24.5	22.0	16.8	9.6	4.2		
	10%	WB	-0.5	-0.3	2.6	6.5	13.1	18.3	20.8	20.2	18.5	12.8	6.9	1.5		
		MCDB	0.4	1.0	4.7	10.3	17.0	21.7	24.3	23.3	20.8	15.0	8.4	2.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.4	9.2	9.7	10.7	10.5	11.0	10.5	10.3	9.1	7.4	7.4		
		MCDBR	7.8	8.5	10.2	13.1	14.0	12.3	12.2	11.3	11.2	10.9	9.3	7.7		
		MCWBR	6.9	6.9	7.3	8.5	8.1	6.9	6.9	6.7	7.2	8.0	7.4	6.8		
		MCDBR	7.4	7.6	9.4	12.1	11.8	10.5	11.0	9.9	9.6	9.4	8.4	7.2		
	5% WB	MCWBR	6.7	6.4	7.1	8.5	7.9	6.5	7.0	6.6	6.7	7.6	7.3	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.276	0.315	0.349	0.384	0.385	0.383	0.380	0.360	0.328	0.298	0.267		
	taud		2.469	2.389	2.386	2.386	2.352	2.390	2.413	2.415	2.465	2.538	2.540	2.496		
	Ebn at Noon		871	912	917	910	884	882	879	869	858	844	813	819		
	Edh at Noon		66	87	102	112	120	116	112	108	95	76	63	59		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.23	3.52	4.58	5.59	6.03	6.19	5.24	3.97	2.39	1.34	0.99		
	RadStd		0.11	0.27	0.33	0.43	0.44	0.47	0.27	0.35	0.29	0.26	0.14	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page